

Electrical Transient Analyzer Program

Short-Circuit Analysis

IEC 60909 Standard

3-Phase, LG, LL, & LLG Fault Currents

	Swing	V-Control	Load	Total			
Number of Buses:	1	1	2	4			
	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	2	0	0	0	0	1	3
	Synchronous Generator	Power Grid	Synchronous Motor	Induction Machines	Lumped Load	Total	
Number of Machines:	0	3	0	0	8	11	

System Frequency:

Unit System:

Project Filename:

Output Filename:

50.00

Metric

substation_design

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Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus2	SWNG	33.000	33.000	1	100.00	0.00
Bus3	Gen.	33.000	33.000	1	100.00	0.00
Bus4	Load	11.000	11.000	1	98.16	-30.00
Bus5	Load	11.000	11.000	1	99.54	-30.00

4 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP .

2-Winding Transformer Input Data

Transformer		Rating				Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	MVA	Prim. kV	Sec. kV	% Z	X/R	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
T1	10.000	33.000	11.000	7.85	156.98	-100.00	-100.00	0	0	0	7.85	Dyn	30.00
T2	10.000	33.000	11.000	7.85	156.98	0	0	0	0	0	7.85	Dyn	30.00

2-Winding Transformer Grounding Input Data

Transformer				Grounding								
				Rating		Conn.	Primary				Secondary	
ID	MVA	Prim. kV	Sec. kV	Type	Type	kV	Amp	ohm	Type	kV	Amp	ohm
T1	10.000	33.000	11.000	D/Y					Solid			
T2	10.000	33.000	11.000	D/Y					Solid			

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA			
ID	Type	From Bus	To Bus	R	X	Z	Y
T1	2W XFMR	Bus3	Bus4	0.50	78.34	78.34	
T2	2W XFMR	Bus3	Bus5	0.50	78.34	78.34	
CB4	Tie Breakr	Bus3	Bus2				

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Power Grid Input Data

Power Grid ID	Connected Bus ID	Rating		% Impedance 100 MVA Base			Grounding Type
		MVASC	kV	R	X"	R/X"	
U1	Bus2	1275.484	33.000	0.05000	7.84000	0.01	Wye - Solid
U2	Bus2	1275.484	33.000	0.05000	7.84000	0.01	Wye - Solid
U3	Bus3	1275.484	33.000	0.05000	7.84000	0.01	Wye - Solid

Total Connected Power Grids (= 3): 3826.452 MVA

Lumped Load Input Data

Lumped Load							Motor Loads								
							Rating		% Load		Loading		% Impedance (Machine Base)		
Lumped Load ID	kVA	kV	Amp	% PF	MTR	STAT	kW	kvar	R	X"	R/X"	Conn.	Type	Amp	MW/PP
Lump1	495.4	11.000	26.00	85.00	80	20	336.87	208.77	2.28	15.21	0.15	Delta			0.34
Lump2	1524.2	11.000	80.00	85.00	80	20	1036.46	642.34	1.53	15.31	0.10	Delta			1.04
Lump3	952.6	11.000	50.00	85.00	80	20	647.77	401.45	2.28	15.21	0.15	Delta			0.65
Lump4	1295.6	11.000	68.00	85.00	80	20	881.01	546.00	2.28	15.21	0.15	Delta			0.88
Lump5	38.1	11.000	2.00	85.00	80	20	25.91	16.06	2.28	15.21	0.15	Delta			0.03
Lump6	38.1	11.000	2.00	85.00	80	20	25.91	16.06	2.28	15.21	0.15	Delta			0.03
Lump7	781.2	11.000	41.00	85.00	80	20	531.22	329.22	2.28	15.21	0.15	Delta			0.53
Lump8	228.6	11.000	12.00	85.00	80	20	155.45	96.34	2.28	15.21	0.15	Delta			0.16

Total Connected Lumped Loads (= 8): 5353.8 kVA

SHORT-CIRCUIT REPORT

Fault at bus: **Bus4**
Nominal kV = 11.000
Voltage c Factor = 1.10 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus4	Total	0.00	8.382	0.00	112.10	115.43	7.836	7.836	1.75E+000	6.89E+001	6.39E+000	8.31E+001
Bus3	Bus4	106.11	7.109	108.17	110.00	108.21	7.043	7.836 *	5.18E-001	8.12E+001	6.39E+000	8.31E+001
Lump1	Bus4	110.00	0.149	110.00	110.00	110.00	0.093	0.000	5.76E+002	3.84E+003		
Lump2	Bus4	110.00	0.458	110.00	110.00	110.00	0.285	0.000	1.26E+002	1.26E+003		
Lump3	Bus4	110.00	0.286	110.00	110.00	110.00	0.178	0.000	2.99E+002	2.00E+003		
Lump4	Bus4	110.00	0.389	110.00	110.00	110.00	0.242	0.000	2.20E+002	1.47E+003		
Bus5	Bus3	106.28	0.004	107.68	109.41	109.44	0.002	0.000	2.52E+004	8.82E+003		
U3	Bus3	110.00	0.789	110.00	110.00	110.00	0.426	0.000	5.50E-002	8.62E+000	8.62E+000	8.62E+000
U1	Bus2	110.00	0.789	110.00	110.00	110.00	0.426	0.000	5.50E-002	8.62E+000	7.16E-001	8.59E+000
U2	Bus2	110.00	0.789	110.00	110.00	110.00	0.426	0.000	5.50E-002	8.62E+000	8.62E+000	8.62E+000
		3-Phase		L-G		L-L		L-L-G				
Initial Symmetrical Current (kA, rms)		:	8.382	:	7.836	:	7.259	:	8.255			
Peak Current (kA), Method C		:	22.901	:	21.411	:	19.833	:	22.557			
Breaking Current (kA, rms, symm)		:		:	7.836	:	7.259	:	8.255			
Steady State Current (kA, rms)		:	7.109	:	7.836	:	7.259	:	8.255			

Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta-Y transformer.

Short-Circuit Summary Report

3-Phase, LG, LL, LLG Fault Currents

Bus		3-Phase Fault			Line-to-Ground Fault				Line-to-Line Fault				*Line-to-Line-to-Ground			
ID	kV	I"k	ip	Ik	I"k	ip	Ib	Ik	I"k	ip	Ib	Ik	I"k	ip	Ib	Ik
Bus4	11.000	8.382	22.901	7.109	7.836	21.411	7.836	7.836	7.259	19.833	7.259	7.259	8.255	22.557	8.255	8.255

All fault currents are in rms kA. Current ip is calculated using Method C.

* LLG fault current is the larger of the two faulted line currents.

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Sequence Impedance Summary Report

Bus		Positive Seq. Imp. (ohm)			Negative Seq. Imp. (ohm)			Zero Seq. Imp. (ohm)			Fault Zf (ohm)		
ID	kV	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance
Bus4	11.000	0.02118	0.83322	0.83349	0.02118	0.83322	0.83349	0.07733	1.00536	1.00833	0.00000	0.00000	0.00000